

1. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix}$, and $C = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 4 \end{bmatrix}$. Calculate each of the following expressions or explain why it is not defined.

a. AB

$$AB = \begin{bmatrix} 1(-1) + 2(0) & 1(1) + 2(1) \\ 3(-1) + 4(0) & 3(1) + 4(1) \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ -3 & 7 \end{bmatrix}$$

b. CB

CB is undefined because the C has 3 columns but B has two rows.

2. Suppose that $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is defined by $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 \\ x_2 + x_3 \end{bmatrix}$.

a. Find the standard matrix A for T .

$$T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \text{ and } T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \text{ So } A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

b. Is T a linear transformation?

$$\text{Yes. You can check the properties or see that } \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_1 \\ x_2 + x_3 \end{bmatrix}.$$

c. Let $\mathbf{x} \in \mathbb{R}^n$ and A be an $m \times n$ matrix. The function $f(\mathbf{x}) = A\mathbf{x}$ is a linear map from $\underline{\mathbb{R}^m}$ to $\underline{\mathbb{R}^n}$.

3. Find A^{-1} or show that it does not exist.

a. $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \rightsquigarrow \text{R2-R3} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \rightsquigarrow \text{R1-R2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$\text{so } A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

b. $A = \begin{bmatrix} 2 & 2 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

A^{-1} does not exist since A is singular.

4. Let $A = \begin{bmatrix} 1/2 & 1/4 \\ 0 & 1/2 \end{bmatrix}$, $A^{-1} = \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

a. Use A^{-1} to solve $A\mathbf{x} = \mathbf{b}$ for $\mathbf{x}$.

$$A\mathbf{x} = \mathbf{b} \Rightarrow \mathbf{x} = A^{-1}\mathbf{b} \text{ so } \mathbf{x} = \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

b. Use your answer from part (a) to express $\mathbf{b}$ as a linear combination of A .

$$1(1/2, 0) + 2(1/4, 1/2) = (1, 1)$$

5. Let $\mathbf{v}_1 = (1, 2)$, $\mathbf{v}_2 = (0, 1)$, and $\mathbf{v}_3 = (1, 0)$ be three vectors in $\mathbb{R}^2$.

a. Show that $\mathbf{v}_1 = (1, 2)$ and $\mathbf{v}_2 = (0, 1)$ gives a basis for $\mathbb{R}^2$.

$$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \rightsquigarrow \text{R2-2R1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ which is nonsingular so it is a basis.}$$

b. Find another basis for $\mathbb{R}^2$.

There are many answers; the standard basis is one of them, $\{(1, 0), (0, 1)\}$.

c. Is $\mathbf{v}_1 = (1, 2)$, $\mathbf{v}_2 = (0, 1)$, and $\mathbf{v}_3 = (1, 0)$ linearly independent? Explain.

No. The $\dim \mathbb{R}^2 = 2$ and this set has three vectors.

d. Does $\mathbf{v}_1 = (1, 2)$, $\mathbf{v}_2 = (0, 1)$, and $\mathbf{v}_3 = (1, 0)$ span $\mathbb{R}^2$?

Yes. Since $\text{Span}\{(1, 2), (0, 1)\} = \mathbb{R}^2$, $(1, 0) \in \text{Span}\{(1, 2), (0, 1)\} \Rightarrow \text{Span}\{(1, 2), (0, 1), (1, 0)\} = \text{Span}\{(1, 2), (0, 1)\} = \mathbb{R}^2$.

6. Let $\mathbf{v}_1 = (2, 4, 8)$, $\mathbf{v}_2 = (-2, -4, -8)$, $\mathbf{v}_3 = (1, 2, 4)$ be vectors in $\mathbb{R}^3$.

a. Is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ linearly independent?

No. $(2, 4, 8) = -1(-2, -4, -8) \Rightarrow (2, 4, 8) - (-2, -4, -8) + 0(1, 2, 4) = \mathbf{0}$. You can also find the reduced echelon form the associated column matrix and see that it is singular (rank of 1).

$$A = \begin{bmatrix} 2 & -2 & 1 \\ 4 & -4 & 2 \\ 8 & -8 & 4 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 2 & -2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

b. Does $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ span $\mathbb{R}^3$?, i.e., does $\text{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = \mathbb{R}^3$? Explain.

No. The $\dim \mathbb{R}^3 = 3$ but $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent (meaning that a basis will have less than 3 vectors).

c. Find the dimension of $\text{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$.

From part (a) we can see that $\text{rank} A = 1$, so a basis will have 1 vector and the dimension must be 1.

d. Find a basis for $\text{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$.

From part (a) we can see that $(2, 4, 8)$ is a basis.

7. Let $V = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{n+1}\}$ be a set of $n + 1$ vectors in $\mathbb{R}^n$.

a. Is V linearly independent or linearly dependent? Explain.

It is linearly dependent. A basis for $\mathbb{R}^n$ will have n vectors ($\dim \mathbb{R}^n = n$). So then a set of $n + 1$ vectors must be linearly dependent.

b. Is V a basis for $\mathbb{R}^n$? Explain.

No. It is linearly dependent.

c. Is $\text{Span} V$ a subspace of $\mathbb{R}^n$?

Yes a span is always a subspace.

Bonus. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ and let $\mathbf{v} \in \mathbb{R}^n$. Prove that $\mathbf{v} \in \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) \iff \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{v})$.

Proof. ($\Rightarrow$) Let $\mathbf{v} \in \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k)$. Then by definition there are real numbers $c_1, c_2, \dots, c_k$ such that $\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k$ so then $\text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{v}) \subset \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k)$. Also $\text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) \subset \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{v})$ so then $\text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{v})$.

($\Leftarrow$) Let $\text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{v})$. $\mathbf{v} \in \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{v})$ so then by the hypothesis $\mathbf{v} \in \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k)$. $\square$